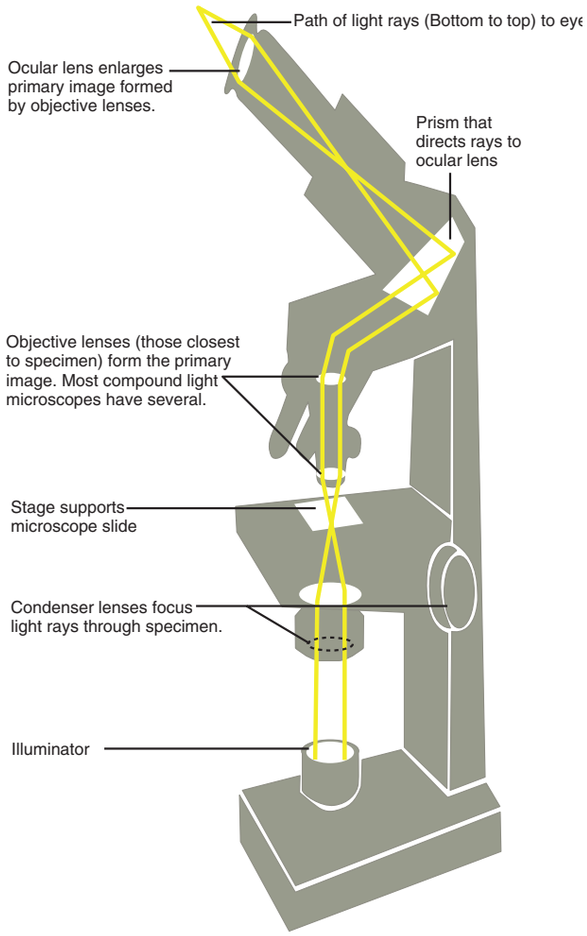


Figure 2.7: The path of light in light Microscopes

Oil Immersion

Oil immersion lens is designed to be in direct contact with oil placed on the cover slip. An oil immersion lens has a short focal length and hence there is a short working distance between the objective lens and the specimen. Immersion oil has a refractive index closer to that of glass than the refractive index of air, so the use of oil increases the cone of light that enters the objective lens. Figure 2.8 explains the working principle of oil immersion objective lens.

HOTS

- What are the two ways by which the resolving power of microscope can be enhanced?
- What are the advantages of the low-power objective over the oil immersion objective for viewing fungi or algae?
- What will happen if water is used instead of immersion oil under a 100X objective lens?

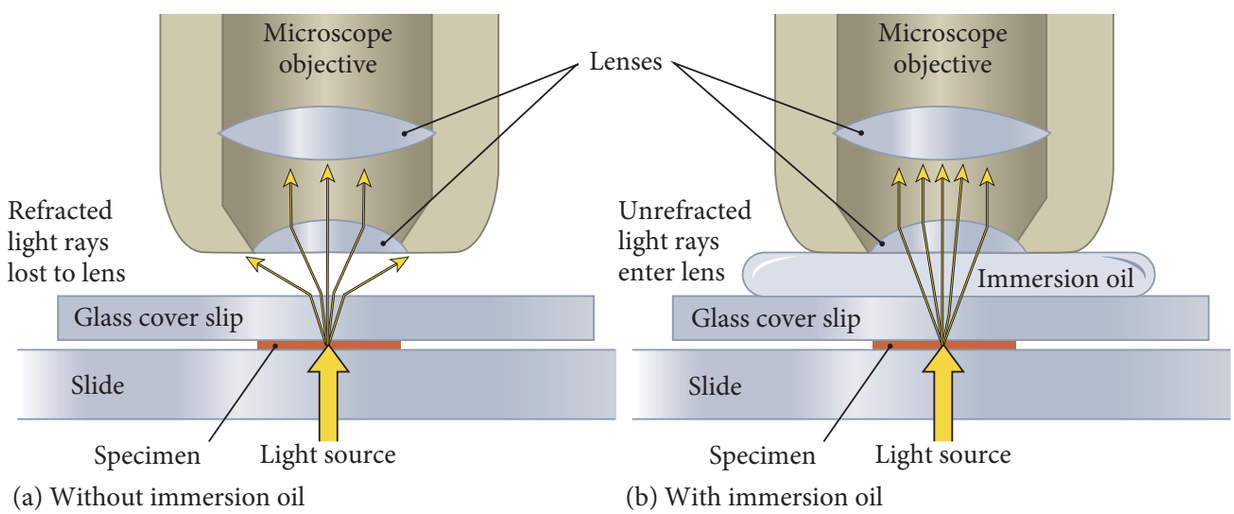


Figure 2.8: Oil Immersion Objective Working Principle

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